SPAT0002

Noise & windows

Maxime Fays. 2023/12/06

Introduction to Signal Processing

- Today's Focus:
  - Solution of Monday’s last exercise
  - PSD parameters
  - Noisy environments and colours
  - Impact of windows
  - Spectrograms

PSD parameters

Nperseg

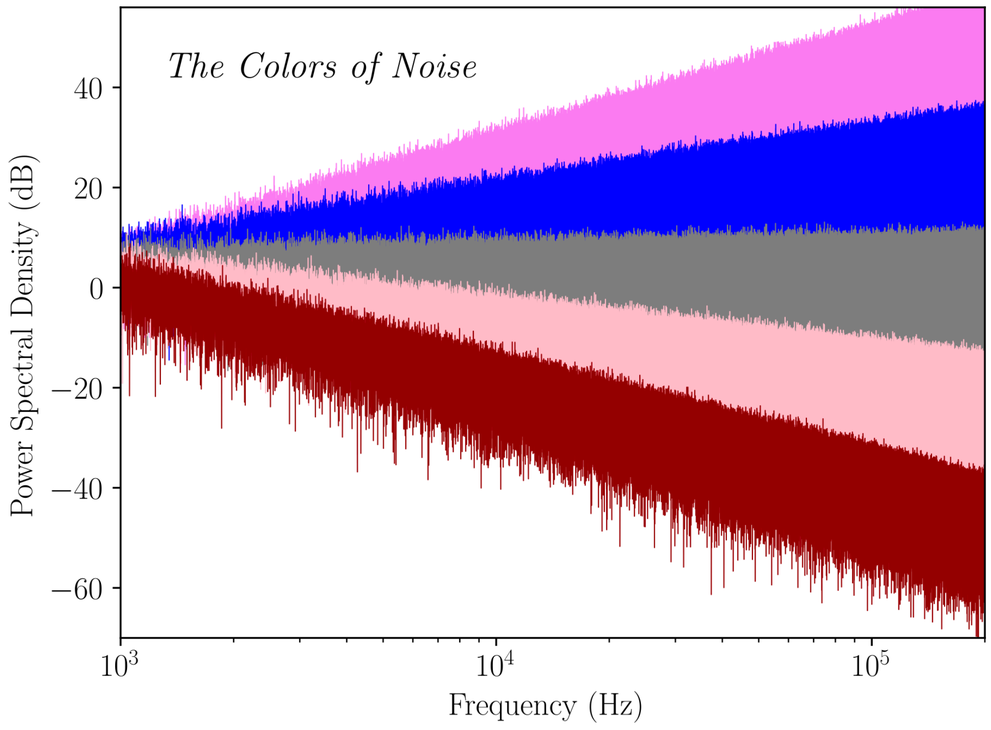
- In Welch's method for PSD estimation, the nperseg parameter plays a crucial role.
- —> specifies the length of each segment of the input data, over which an individual periodogram is computed.
- Increased Resolution in Frequency Domain:
  - Increasing nperseg results in longer segments, which leads to higher frequency resolution in the PSD estimate
  - —> finer details in the frequency domain.
  - —> reduced frequency averaging, which might increase the variance of the PSD estimate.
- Improved Averaging, Less Variance:
  - Decreasing nperseg results in shorter segments. This leads to more segments over which to average, reducing the variance of the PSD estimate.
  - —> Lower frequency resolution: fine details in the frequency domain might be lost or smeared out.
- Frequency Bins: A larger nperseg will produce more frequency bins, giving a more detailed frequency spectrum. Conversely, a smaller nperseg results in fewer frequency bins and a coarser frequency spectrum.

Noisy environments

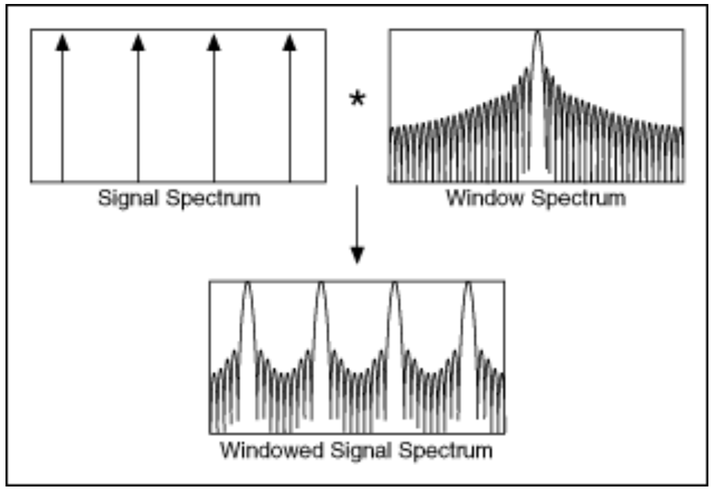
- **Increased Power Level**: Noise adds to the overall power across the PSD spectrum.
- **Feature Masking**: Key signal features can be obscured by dominant noise frequencies.
- **Spectral Shape Alteration**: Specific noise profiles (e.g., pink or blue noise) can change the apparent frequency distribution of the signal.
- **Spurious Peaks**: Noise may introduce false peaks, not present in the original signal.
- **Broadening of Spectral Lines**: Sharp signal features can become less distinct due to noise.
- **Reduced Signal-to-Noise Ratio**: Overall distinction between signal and noise diminishes, affecting analysis quality.

Coloured noise

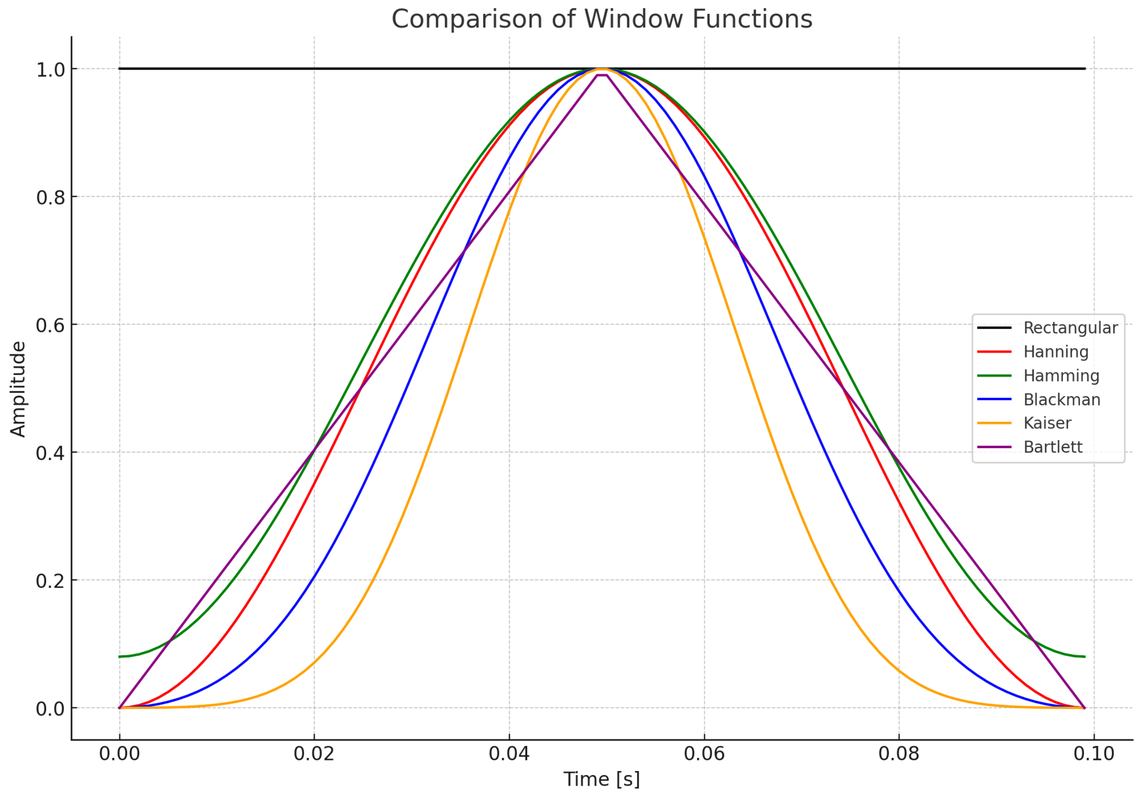
- Flat Spectrum Noise
  - White Noise: Uniform intensity across all frequencies.
    - Applications: Audio testing, electronic calibration, statistical analysis.
- Low-Frequency Emphasized Noise
  - Pink Noise: Power decreases with frequency increase (-3 dB/octave).
  - Brown Noise: Power decreases more sharply with frequency (-6 dB/octave).
    - Applications: Natural sound simulation, acoustics, sleep aids, psychoacoustic testing.
- High-Frequency Emphasized Noise
  - Blue Noise: Power increases with frequency.
  - Violet Noise: Steeper power increase with frequency (+6 dB/octave).
    - Applications: Audio dithering, high-frequency applications, spatial processing.
- Perception-Based Noise
  - Grey Noise: Perceived as even by the human ear, modeled on equal-loudness contours.
  - Applications: Audio engineering, sound masking, hearing tests.
  - Green Noise: Associated with natural ambient sounds like rustling leaves.
    - Applications: Relaxation, sound therapy, natural soundscapes.



Windows in signal processing

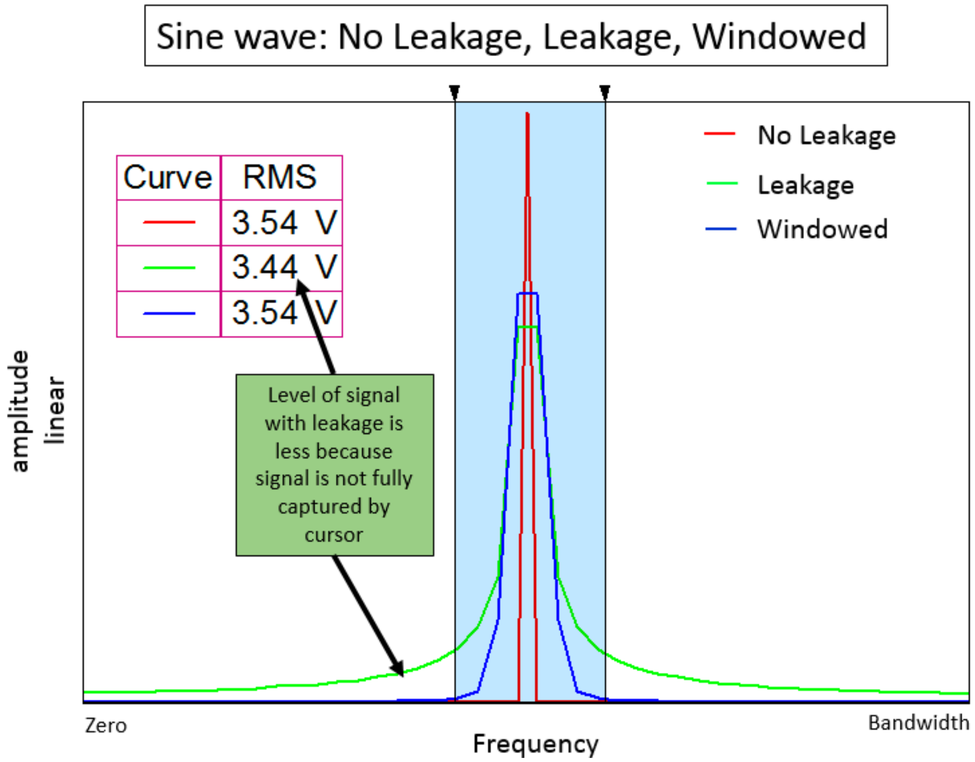
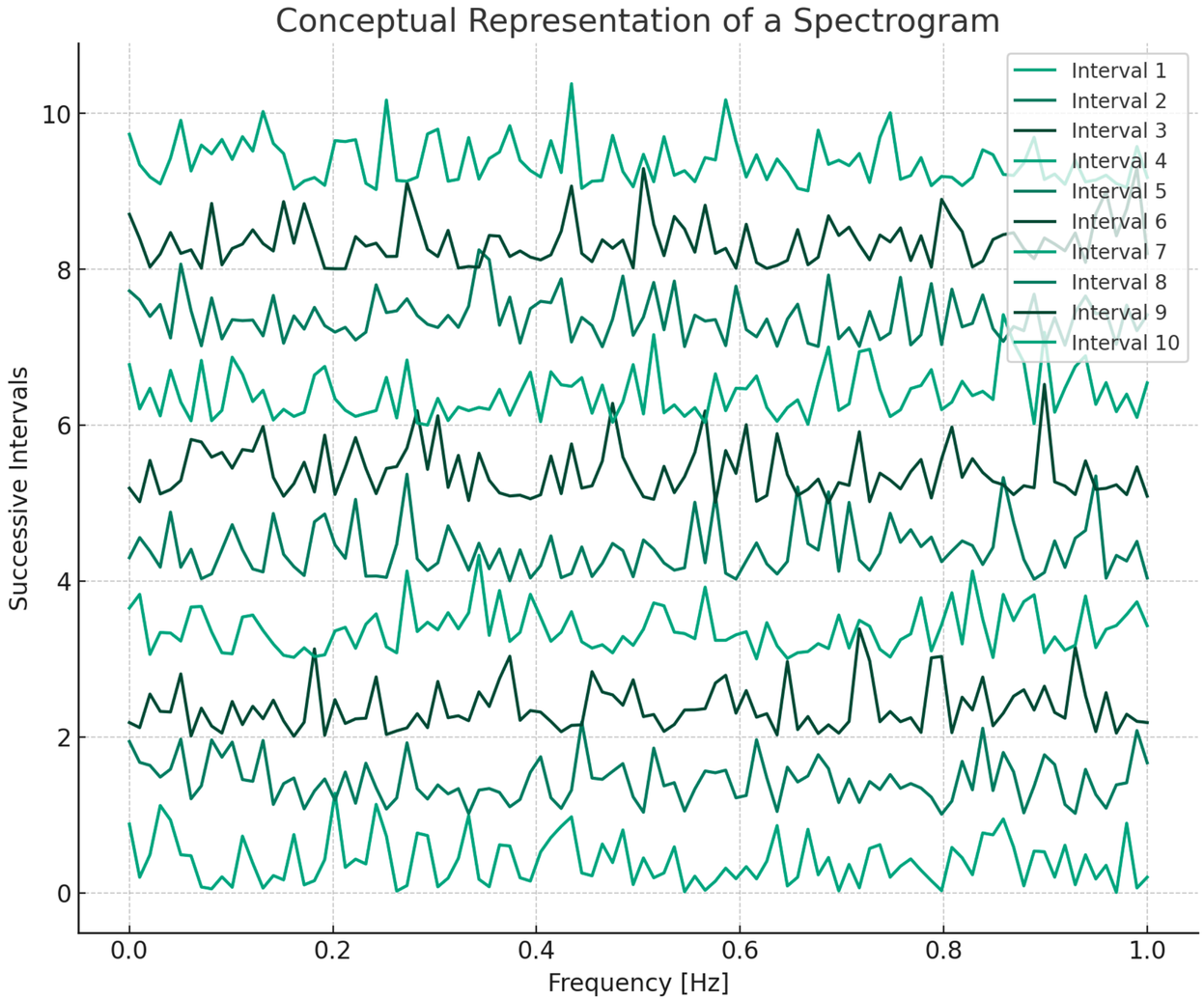
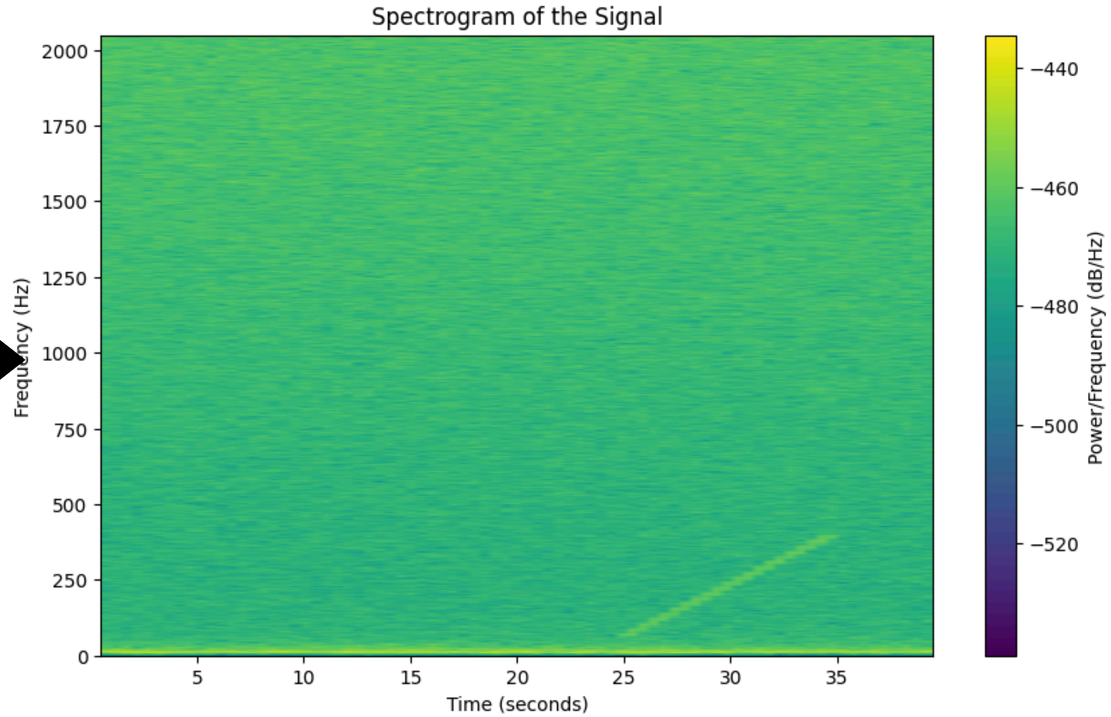


- Purpose of Windowing: Minimizes spectral leakage by tapering the signal at the edges.
- Rectangular Window:
  - Basic, no windowing effect.
  - High spectral leakage.
  - Best frequency resolution.
- Hanning and Hamming Windows:
  - Reduces spectral leakage better than rectangular.
  - Balances frequency resolution and leakage.
  - Used in general-purpose signal processing.
- Blackman Window:
  - Further reduces spectral leakage.
  - Lower frequency resolution.
  - Ideal for signals with close frequency components.



Spectral leakage

- Effects on PSD:
  - Peak Broadening: Non-rectangular windows broaden spectral peaks.
  - Amplitude Reduction: Windowing reduces peak amplitudes.
- Spectral Leakage Reduction:
  - Non-rectangular windows mitigate leakage, preventing false frequency components in PSD.
- Choice of Window:
  - **<u>Depends on signal characteristics and analysis requirements.</u>**



Spectrograms